

BROHOVO, Russian Federation

WMO: 259270

Lat: 59.638N

Lon: 154.303E

Elev: 18

StdP: 14.69

Time Zone: 11.00 (E11)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-27.4	-24.1	-32.4	0.9	-26.1	-29.5	1.1	-23.6	46.8	20.4	41.0	18.4	5.2	320	0.937

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.8	69.2	59.7	65.6	58.2	62.8	56.9	61.0	67.2	59.2	64.2	57.7	61.9	12.7	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
58.3	72.7	63.8	56.7	68.6	61.7	55.4	65.4	59.9	27.1	67.5	25.9	64.6	25.0	62.3	66.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 41.4	35.4	29.9	DB	-31.3	77.2	2.3	2.9	-32.9	79.2	-34.3	80.9	-35.5	82.5	-37.2	84.6	(4)
(5)			WB	-31.5	64.4	2.2	1.7	-33.1	65.7	-34.4	66.7	-35.6	67.7	-37.2	69.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	26.3	-1.5	-3.0	5.4	19.2	33.2	46.6	56.6	55.9	47.4	32.0	15.2	6.1		
	DBStd	23.21	14.35	14.36	13.17	10.23	4.61	5.79	4.31	3.50	5.30	6.95	10.24	12.11		
	HDD50	9114	1597	1484	1381	925	522	133	2	1	107	557	1043	1362		
	HDD65	14141	2062	1904	1846	1375	987	552	263	283	527	1022	1493	1827		
	CDD50	450	0	0	0	0	0	30	206	184	30	0	0	0		
	CDD65	4	0	0	0	0	0	0	3	1	0	0	0	0		
	CDH74	25	0	0	0	0	0	4	19	2	0	0	0	0		
	CDH80	1	0	0	0	0	0	0	1	0	0	0	0	0		
	WSAvg	12.7	12.3	11.8	12.6	11.4	9.7	10.4	11.6	12.0	13.1	14.5	16.5	16.6		
	PrecAvg	22.2	1.4	1.5	1.2	1.2	1.0	1.2	2.1	2.2	2.4	2.8	2.6	1.9		
(15) Precipitation	PrecMax	34.1	3.0	4.3	3.9	3.4	3.2	4.2	6.9	5.3	5.8	6.2	5.9	7.2		
	PrecMin	11.8	0.1	0.1	0.1	0.1	0.0	0.0	0.0	0.3	0.1	0.4	0.2	0.2		
	PrecStd	5.3	0.9	1.1	0.9	0.9	0.8	1.0	1.5	1.3	1.4	1.5	1.4	1.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.5	32.6	30.5	37.4	53.1	67.8	76.6	71.3	63.4	47.7	39.2	33.2		
		MCWB	30.4	31.3	29.7	32.8	44.5	55.6	61.8	60.8	52.7	43.8	37.2	32.3		
	2%	DB	28.9	28.9	28.1	34.2	47.4	62.2	71.3	67.5	59.5	46.2	37.1	31.2		
		MCWB	28.4	28.1	27.2	31.8	41.4	53.0	61.3	59.4	52.4	42.3	35.9	30.3		
	5%	DB	26.0	23.9	26.2	33.0	43.3	58.9	67.6	64.3	56.7	43.1	32.4	28.1		
		MCWB	25.2	23.0	25.2	31.3	38.9	51.3	59.1	57.8	51.6	39.4	31.8	27.0		
	10%	DB	22.0	18.9	23.7	31.7	40.4	55.8	64.1	61.9	54.5	40.8	29.9	24.4		
		MCWB	21.1	18.0	22.9	30.5	36.8	50.6	58.3	56.8	50.6	38.2	29.1	23.4		
	0.4%	WB	30.6	31.4	30.1	33.7	45.3	57.3	63.9	63.3	55.9	45.1	37.5	32.6		
		MCDB	31.3	32.5	30.4	35.2	52.3	65.5	72.6	68.9	59.8	47.1	38.8	32.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	28.5	28.2	27.4	32.6	41.6	54.0	61.9	60.5	54.1	42.7	35.8	30.4		
		MCDB	28.8	28.7	28.1	33.8	46.7	59.7	68.7	65.7	56.7	45.4	37.4	31.3		
	5%	WB	25.3	23.2	25.5	31.6	39.2	52.3	60.2	58.7	52.8	40.1	31.8	27.0		
		MCDB	25.8	23.8	26.3	32.8	42.9	57.4	66.1	62.9	55.2	42.3	32.2	27.9		
	10%	WB	21.0	17.9	22.9	30.5	37.1	50.8	58.6	57.4	51.6	38.6	29.2	23.7		
		MCDB	22.0	18.9	23.6	31.6	39.8	55.4	63.3	61.1	54.0	40.6	29.9	24.4		
	5% DB	MDBR	11.2	11.6	14.4	14.7	10.2	11.0	10.8	9.4	9.6	8.7	9.9	9.5		
		MCDBR	9.9	8.5	10.0	12.4	17.0	18.7	18.5	15.2	13.0	9.4	8.9	8.6		
		MCWBR	9.6	8.9	9.8	10.9	11.2	10.5	9.8	9.0	8.0	7.5	9.1	8.5		
		MCWBR	9.6	8.6	10.0	10.4	15.7	16.1	16.0	12.3	10.1	7.5	9.0	8.7		
(35) Mean Daily Temperature Range	5% WB	MCWBR	9.8	9.0	9.6	9.2	10.6	9.7	9.4	8.2	7.1	6.9	9.2	8.7		
(40) Clear-Sky Solar Irradiance	taub		0.238	0.277	0.332	0.407	0.437	0.407	0.455	0.423	0.346	0.323	0.273	0.235		
	taud		2.055	2.082	1.970	1.815	1.941	2.284	2.166	2.251	2.445	2.278	2.088	1.978		
	Ebn at Noon		172	227	245	244	248	262	243	242	248	209	151	120		
	Edh at Noon		15	25	39	55	53	38	42	36	25	21	14	11		
(44) All-Sky Solar Radiation	RadAvg		126	397	898	1422	1657	1723	1639	1207	824	409	140	61		
	RadStd		18	49	94	137	144	177	134	151	102	36	14	5		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page